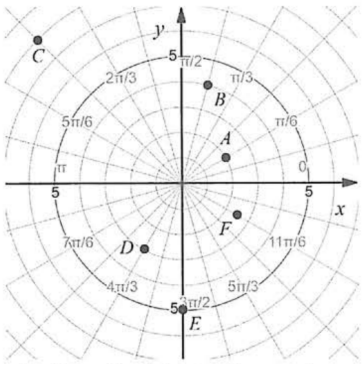
Polar Coordinates: Introduction

Questions 1–11

Question 1.



a) Identify each of the 6 points on the polar coordinate grid above as polar coordinates, giving your answers in the form (r, θ) with $r > 0$ and $-\pi < \theta \leq \pi$
b) State the point furthest from the pole O
c) State the point closest to the pole O

Question 2. Given each of the following points are polar coordinates, plot each on the same grid:

a) $A(2, \frac{\pi}{2})$
b) $B(5, \frac{4\pi}{3})$
c) $C(3, \frac{\pi}{3})$
d) $D(2, -\frac{\pi}{3})$
e) $E(4, -\frac{7\pi}{6})$

Question 3. The following points are given as Cartesian coordinates. Find the polar form of each of the points giving your answer in the form (r, θ) where r and θ are both exact:

a) $(3, 3)$
b) $(5, -5)$
c) $(2, 2\sqrt{3})$
d) $(-1, -\sqrt{3})$
e) $(-3, \sqrt{3})$

Question 4. Find the principal polar coordinate of each of the following points given in Cartesian form giving your answer in the form (r, θ) where r is exact and θ is given to 3 significant figures:

a) $(1, 3)$
b) $(-2, 5)$
c) $(-7, -24)$
d) $(12, -13)$

Question 5. Convert each of the following polar coordinates into principal polar coordinates:

a) $A(2, \frac{5\pi}{4})$
b) $B(1, \frac{11\pi}{6})$
c) $C(-1, \frac{\pi}{6})$
d) $D(-2, \frac{4\pi}{3})$
e) Plot each of the 4 points in parts a) to d) on a polar coordinate grid.

Question 6. Convert each of the following polar coordinates into Cartesian form:

a) $A(3, \frac{\pi}{4})$
b) $B(2, \frac{5\pi}{6})$
c) $C(-4, \frac{\pi}{3})$

Question 7. A principal polar coordinate is given by $(3, \frac{\pi}{3})$
Two points $(-3, p)$ and $(3, q)$ are known to be the same point as $(3, \frac{\pi}{3})$
Find the value of p and q such that $(-3, p)$ and $(3, q)$ where $0 < p < 2\pi$ and $2\pi < q < 4\pi$

Question 8. An equilateral triangle ABC has a vertex with polar coordinates $A(2, \frac{\pi}{6})$. Find the possible set of polar coordinates of ABC when:

a) the pole O is at the centre of the triangle.
b) vertex B is at the pole.

Question 9. The points A, B and C have polar coordinates $(3, \frac{\pi}{4})$, $(4, \frac{3\pi}{4})$ and $(4, \frac{17\pi}{12})$ respectively. Find the area of the triangle ABC

Question 10. A curve C has polar equation given by

$$r = 4 - 3\sin\theta \text{ for } -\pi < \theta \leq \pi$$

a) Verify that the point given by polar coordinates $(1, \frac{\pi}{2})$ lies on C
b) Find the point on the curve which is furthest away from the pole O
c) Find the value of the cartesian coordinates of the point which lies on the curve where $\theta = \frac{\pi}{4}$ giving your answer in exact form.

Question 11. Given that a and b are positive constants, find the conditions under which polar coordinates $(-a, b)$ and $(a, -b)$ describe the same point.